Renwei Meng

Mobile: (+86) 182-3513-0053 | Email: r32314095@stu.ahu.edu.cn | Github: renweimeng | Homepage: renweimeng.com
Google Scholar | ORCID: 0009-0001-6879-6629 | Undergraduate Student, Anhui University

EDUCATION

Anhui University Hefei, China
Bachelor of Engineering in Digital Media Technology Sep. 2023 – Jun. 2027

- **GPA:** 3.95 / 4.0, 4.34 / 5.0, 92.58/100 **(Ranking: 1/100)**
- **IELTS:** 7.0/9.0
- **Core Courses:** Diffusion & Flow Matching, Machine Learning, Mathmatical Foundation of Reinforcement Learning, Data Structures, Computer Organization, Operating Systems, Computer Networks, Linear Algebra, Probability Theory, Digital Image Processing, Computer Graphics, Python.

RESEARCH INTERESTS

- **Trustworthy Generative AI:** Manifold-aware evaluation, reward hacking detection, reinforcement learning, preference optimization, and flow-based generative modeling.
- **Reliable LLMs and Agentic AI:** Knowledge-boundary evaluation, epistemic uncertainty and abstention, retrieval-augmented generation, multi-agent systems, and evidence-grounded code auditing.
- **Multimodal and Neurocognitive Intelligence:** EEG-based affective computing, cross-subject neural decoding, brain-computer interfaces, gaze modeling, and multimodal detection of AI-generated content.

PUBLICATIONS & PATENTS

Publications

- **Normal Fréchet Drift: A Geometric Measure for Detecting Off-Manifold Deviation in Generative Models**
  Renwei Meng*, Yansen Han*, Tao Lin *(co-first authors; equal contribution)*
  Submitted to International Conference on Learning Representations (ICLR), 2027. **Under Review**
  - Proposed Normal Fréchet Drift (NFD), a geometry-aware metric that quantifies normal deviation from pretrained data manifolds via local tangent patches, with theoretical guarantees and systematic validation on controlled manifolds and text-to-image models.
- **MORPH: Bi-Temporal Online Reinforcement Learning for Structure-Aware Music Flow Matching**
  Renwei Meng
  Submitted to the AAAI Conference on Artificial Intelligence (AAAI), 2027. **Under Review**
  - Proposed MORPH, a bi-temporal online reinforcement learning framework that performs counterfactual credit assignment jointly over flow steps and musical sections, enabling structure-aware preference optimization for long-form music generation.
- **Know2Guess: A Contamination-Aware Multi-Zone Benchmark for Knowledge-Boundary Evaluation in Large Language Models**
  Renwei Meng, Bowen Zhang, Jian Wang, et al.
  Accepted by International Conference on Neural Information Processing (ICONIP), 2026. **Accepted**
  - Introduced a 1,200-item, contamination-aware benchmark with four knowledge zones to jointly evaluate factual answering, epistemic abstention, policy refusal, prompt sensitivity, and knowledge-boundary reliability across nine open-weight LLMs.
- **Conditional Riemannian Adaptive Flow Transport for Cross-Subject EEG Decoding**
  Renwei Meng, Hongyi Zhang
  Accepted by the 23rd Pacific Rim International Conference on Artificial Intelligence (PRICAI), 2026. **Accepted**
  - Proposed CRAFT, a conditional Riemannian adaptive flow transport framework combining SPD-manifold alignment, neural residual flows, and label-free test-time adaptation for robust cross-subject EEG decoding across five public benchmarks.
- **Group Resonance Network (GRN) for Cross-Subject EEG Emotion Recognition**
  Renwei Meng
  Accepted by International Conference on Artificial Neural Networks (ICANN), 2026. **Accepted**
  - Proposed a group-aware EEG emotion recognition framework integrating learnable prototypes and multi-subject resonance modeling via PLV/coherence synchrony to improve cross-subject generalization on SEED and DEAP benchmarks.
- **S-AODP++: Structure-Aware AOI-Guided Scanpath Attention for AI-Generated Image Identification**
  Jingming Wang*, **Renwei Meng***, Yunjie Si* *(co-first authors; equal contribution)*
  Accepted by the 28th ACM International Conference on Multimodal Interaction (ICMI), 2026. **Accepted**
  - Proposed S-AODP++, a gaze-supervised multimodal framework that maps fixations to structure-aware AOIs and

combines a gaze Transformer, AOI transition graph, and semantic-aware scanpath attention for eye-tracker-free AI-generated image identification, achieving 81.7% accuracy and 85.0% AUROC.

- **Explainable Innovation Engine: Dual-Tree Agent-RAG with Methods-as-Nodes and Verifiable Write-Back**
  Renwei Meng **Arxiv**
  - Proposed a dual-tree Agent-RAG framework that enables explainable method-level retrieval and controllable innovation through structured synthesis and verification.
- **Breaking the Support Barrier in RLVR: A Fork-Conditional Reachability Criterion for Reliable Boundary Expansion**
  Renwei Meng
  Submitted to The 18th Asian Conference on Machine Learning (ACML), 2026. **Under Review**
  - Proposed FCR-PO, a self-contained fork-conditional exploration framework for RLVR that predicts when reliable support expansion is feasible and uses verifier-gated branch injection with clipped off-policy optimization to improve reasoning coverage at substantially lower compute than tree search.
- **VulnAgent-R2: Evidence-Calibrated Multi-Agent Auditing for Repository-Level Vulnerability Detection**
  Renwei Meng*, Haoyi Wu*, Jingming Wang* *(co-first authors; equal contribution)*
  Submitted to The 18th Asian Conference on Machine Learning (ACML), 2026. **Under Review**
  - Designed a multi-stage agentic vulnerability detection pipeline combining risk screening, repository context retrieval, specialised analysis agents, and evidence fusion to improve detection accuracy and localisation on realistic benchmarks.

Patents

- **A generative deepfake video detection method and its system**
  Shiang Hu, Zhiwen Zha, **Renwei Meng**, Yuhao Fang.
  China Invention Patent, Application No. 202610165744.1. **Accepted**
- **Resting-state EEG quality assessment method and system based on dual-branch contrastive learning**
  Shiang Hu, Dongdong Jia, Wan'er Lv, **Renwei Meng**, Chao Lv.
  China Invention Patent, Application No. 202512044739.3. **Accepted**

RESEARCH EXPERIENCE

Manifold-Aware Detection of Reward Hacking in Generative Models (NFD) Jul. 2026 – Present
Core Researcher | Advisor: Prof. Tao Lin | Westlake University

- **Problem:** Investigated reward hacking in preference-optimized generative models, where reward scores improve while generated samples drift away from the pretrained data manifold; conventional metrics such as FID, KL divergence, and reward scores cannot reliably isolate this off-manifold degradation.
- **Method:** Proposed **Normal Fréchet Drift (NFD)**, a manifold-aware metric that reconstructs local geometry with multi-patch tangent estimators and separates tangential variation from harmful normal drift; established theoretical guarantees and developed an oracle-free hyperparameter selection procedure.
- **Result:** Validated NFD through analytical manifold benchmarks, estimator ablations, real text-to-image checkpoints, and blind human evaluation, demonstrating that it identifies visual degradation missed by reward, FID, nearest-neighbor, and divergence-based baselines. The resulting manuscript is under review at **ICLR 2027**.

Research on Inter-brain Synchrony for AI-Generated Content Detection (EEG Hyperscanning) Sep. 2025 – Sep. 2026
Core Researcher | Advisor: Prof. Shiang Hu

- **Problem:** Addressed the growing challenge of distinguishing AI-generated images/videos and detecting hallucinations in LLM outputs, which remain difficult for traditional computer vision and rule-based detection methods.
- **Method:** Conducted dual-brain EEG hyperscanning experiments and developed a neural-cognition-inspired framework integrating multi-band EEG features with advanced deep models, including Transformer-based multimodal encoders, Graph Neural Networks (GNN), and attention-based cross-brain feature fusion.
- **Result:** The proposed model achieved **93.7% accuracy** and **92.9% F1-score**, outperforming strong baselines such as CNN, SVM, and standard multimodal detectors.

Industrial Metaverse Intelligent Interaction System (Multi-modal) Sep. 2025 – Jun. 2026
Algorithm Engineer | Research Group | Advisor: Prof. Shiang Hu

- Developing a real-time interaction system integrating **Eye-tracking**, **Gesture Recognition**, and **Pose Estimation**.
- Designed a multi-modal fusion algorithm to interpret user intent in industrial scenarios with low latency.
- Optimized 3D rendering modules and data transmission protocols to ensure smooth interaction in the metaverse environment.

Software Vulnerability Detection based on Transformer Architecture Mar. 2025 – Jan. 2026
Project Leader | National College Student Innovation Program | Advisor: Assoc. Prof. Cheng Zhang | Anhui University

- Developing a semantic-aware vulnerability detection model using **Transformer** architecture to address the limitations of RNNs in capturing long-range dependencies in code.
- Utilizing **Self-Attention mechanisms** to extract deep semantic features from Abstract Syntax Trees (AST) and Control Flow Graphs (CFG) of source code.
- Curating a large-scale dataset of C/C++ vulnerabilities and evaluating the model's precision and recall against static analysis tools.

Coronary CTA Image Artifact Removal and Clarity Reconstruction Jan. 2025 – Jun. 2025
Research Assistant | Advisor: Prof. Zhe Jin | Anhui University

- Focused on removing motion artifacts and noise in Coronary CT Angiography (CTA) to enhance clinical diagnostic quality.
- Designed a deep learning denoising framework combining **U-Net** with **Generative Adversarial Networks (GAN)**.
- Implemented the pipeline in PyTorch. The proposed method significantly improved the Signal-to-Noise Ratio (SNR) and Structural Similarity Index (SSIM) compared to traditional filtering methods (e.g., BM3D).

Regional Carbon Emission Price Prediction (CNN-GRU-LSTM-Attention) Jun. 2024 – Jun. 2025
Project Leader | National College Student Innovation Program | Advisor: Assoc. Prof. Zhanghangjian Chen | Anhui University

- Proposed a hybrid time-series prediction model combining **CNN** (local feature extraction), **GRU/LSTM** (temporal dependency), and **Attention mechanisms**.
- Validated on data from 7 major carbon trading markets in China, achieving higher prediction accuracy and robustness compared to single-structure models.

INTERNSHIP EXPERIENCE

Taiyuan Longwei Electronic Technology Co., Ltd. Taiyuan, China
Java Backend Developer Intern Jul. 2025 – Sep. 2025

- Developed robust backend interfaces and managed database operations using **Spring Boot** and **MyBatis** frameworks.
- Collaborated with the frontend team to design and implement RESTful APIs, optimizing data transmission efficiency and system stability.

State Grid Shanxi Electric Power Co., Ltd. Taiyuan, China
System Operations and Maintenance Intern Jul. 2024 – Sep. 2024

- Maintained the stability and security of the power operation system platform, ensuring high availability for critical grid management tasks.
- Performed data analysis on operational metrics and generated comprehensive monthly reports to support data-driven decision-making for system optimization.

HONORS & AWARDS

- **National Second Prize**, National College Students Market Survey and Analysis Competition 2026
- **Finalist**, Mathematical Contest in Modeling (MCM/ICM) 2026
- **National Scholarship** (Top 0.1%, Highest Honor for Undergraduates in China) 2025
- **First Prize**, National College Student Mathematical Modeling Competition (Provincial) 2025
- **Second Prize**, Lanqiao Cup National Software Talent Competition (Java, Provincial) 2025
- **Honorable Mention**, Mathematical Contest in Modeling (MCM/ICM) 2025
- **University First-Class Academic Scholarship** 2024, 2025

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, C/C++, Java, R, SQL, HTML, Bash, zsh.
- **Frameworks & Tools:** PyTorch, TensorFlow, Git, Docker, Linux, E-Prime 3.
- **Typesetting & Design:** LaTeX, Visio, Adobe Photoshop, Adobe Illustrater, PowerPoint, Figma.
- **AI tools:** ChatGPT, Codex, Claude Code, n8n, LangChain, LangGraph, ComfyUI.